

**LIST OF OIL SENSITIVE AREAS**

[Updated as at Sep 2011]

| S/N                  | FACILITIES                             |                                                 | AGENCY-<br>IN-CHARGE | IMPACT ON<br>PUBLIC<br>SAFETY | ECONOMIC<br>IMPACT | IMMEDIACY<br>OF IMPACT | PUBLIC<br>ASSURANCE | TOTAL<br>SCORE | CONSIDERATIONS FOR PRIORITISATION<br>IN PROTECTION                                                                               |
|----------------------|----------------------------------------|-------------------------------------------------|----------------------|-------------------------------|--------------------|------------------------|---------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------|
|                      | NAME                                   | TYPE                                            |                      |                               |                    |                        |                     |                |                                                                                                                                  |
| Cat 1 (Top Priority) |                                        |                                                 |                      |                               |                    |                        |                     |                |                                                                                                                                  |
| 1                    | Marina Barrage                         | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 2                    | Jurong Tidal Gates                     | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 3                    | Kranji Tidal Gates                     | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 4                    | Lower Seletar Tidal Gates              | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 5                    | Pandan Tidal Gates                     | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 6                    | Poyan Tidal Gates                      | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 7                    | Punggol Tidal Gates                    | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 8                    | Serangoon Tidal Gates                  | Tidal Gates                                     | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Affect operation of tide gates.                                                                                                  |
| 9                    | SingSpring Desalination Plant          | Desalination Plants and Variable Salinity Plant | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Oil will disrupt water supply operations.                                                                                        |
| 10                   | Tuas Desalination Plant                | Desalination Plants and Variable Salinity Plant | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Oil will disrupt water supply operations.                                                                                        |
| 11                   | Variable Salinity Plant (At Pasir Ris) | Desalination Plants and Variable Salinity Plant | PUB                  | 3                             | 2                  | 2                      | 3                   | 10             | Oil will disrupt water supply operations.                                                                                        |
| 12                   | Senoko Power Station (Senoko)          | Power Stations                                  | EMA                  | 3                             | 3                  | 2                      | 2                   | 10             | Generates approximately 28% of Singapore's electricity supply. Power station has inflated boom to deal with oil spill incidents. |
| 13                   | Seraya Power Station (Jurong Island)   | Power Stations                                  | EMA/JTC              | 3                             | 3                  | 2                      | 2                   | 10             | Generates approximately 25% of Singapore's electricity supply. Power station has inflated boom to deal with oil spill incidents. |
| 14                   | Tuas Power Station (Tuas)              | Power Stations                                  | EMA                  | 3                             | 3                  | 2                      | 2                   | 10             | Generates approximately 24% of Singapore's electricity supply. Power station has inflated boom to deal with oil spill incidents. |

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|------------------------------|---------------------------------------------------------------------------------------------------|---------------------------|----------------------|-------------------------------|--------------------|------------------------|---------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | NAME                                                                                              | TYPE                      |                      |                               |                    |                        |                     |                |                                                                                                                                                                                                                               |
| 15                           | Sembcorp Cogen<br>(Jurong Island)                                                                 | Power Stations            | EMA/JTC              | 3                             | 3                  | 2                      | 2                   | 10             | Generates approximately 11% of Singapore's electricity supply. Power station has inflated boom to deal with oil spill incidents. 1x 400 MW CCGT (CCP 3) to commence commercial operation in 2014/15 at Banyan, Jurong Island. |
| 16                           | Keppel Merlimau Cogen (Jurong Island)                                                             | Power Stations            | EMA/JTC              | 3                             | 3                  | 2                      | 2                   | 10             | Generates approximately 9% of Singapore's electricity supply. Power station has inflated boom to deal with oil spill incidents.                                                                                               |
| 17                           | Island Power Coy<br>(Now known as GMR Energy)                                                     | Power Stations            | EMA                  | 3                             | 3                  | 2                      | 2                   | 10             | IPC owns 796MW two-unit CCGT to commence commercial operation in 2013/14 at Jurong Island Highway, Jurong Island.                                                                                                             |
| 18                           | Fish farming area at Southern Islands<br>(Off Pulau Semakau)                                      | Fish Farms                | SFA                  | 2                             | 1                  | 2                      | 3                   | 8              | Major fish farming site for local supply of fish. Ensure production not affected.                                                                                                                                             |
| 19                           | Fish farming area at West Johor Strait<br>(Off Lim Chu Kang)                                      | Fish Farms                | SFA                  | 2                             | 1                  | 2                      | 3                   | 8              | Important fish farming site. Ensure production not affected.                                                                                                                                                                  |
| 20                           | Fish farming areas at East Johor Strait<br>(Off Pulau Ubin)                                       | Fish Farms                | SFA                  | 2                             | 1                  | 2                      | 3                   | 8              | Important fish farming site. Ensure production not affected.                                                                                                                                                                  |
| 21                           | Fish farming areas at East Johor Strait<br>(Off Pulau Tekong)                                     | Fish Farms                | SFA                  | 2                             | 1                  | 2                      | 3                   | 8              | Important fish farming area contributing to local supply.                                                                                                                                                                     |
| 22                           | Fish farming areas at East Johor Strait<br>(Serangoon Harbour / Pasir Ris )                       | Fish Farms                | SFA                  | 2                             | 1                  | 2                      | 3                   | 8              | Considered together with P Ubin fish farming sites as one cluster.                                                                                                                                                            |
| 23                           | SFA Fish Farm off St John's Island<br>(Cooper Channel)                                            | Fish Farms                | SFA                  | 2                             | 2                  | 2                      | 1                   | 7              | SFA 's experimental fish research farm. Ensure research is not affected.                                                                                                                                                      |
| 24                           | SFA Marine Aquaculture Centre on St John's Island<br>(seawater intake at Sisters Fairway)         | Fish Farms                | SFA                  | 2                             | 2                  | 2                      | 1                   | 7              | SFA 's and TMSI's marine research center. Ensure research is not affected.                                                                                                                                                    |
| <b>Cat 2 (High Priority)</b> |                                                                                                   |                           |                      |                               |                    |                        |                     |                |                                                                                                                                                                                                                               |
| 25                           | East Coast Park<br>(includes Cableski 360 facility, breakwaters, revetments, seawalls and bridge) | Park and Recreation Areas | NParks, NEA, BCA     | 2                             | 1                  | 1                      | 3                   | 7              | Largest stretch of recreational park by the beach. Public recreational beach will have to be closed and cleaned up                                                                                                            |

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|-----|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------|-------------------------------|--------------------|------------------------|---------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | NAME                                                                                         | TYPE                                                                       |                       |                               |                    |                        |                     |                |                                                                                                                                                                                                                                                                    |
| 26  | Public beaches in Sentosa: Siloso beach (and headlands), Palawan beach, Tanjong Beach        | Park and Recreation Areas                                                  | SDC                   | 2                             | 1                  | 1                      | 3                   | 7              | If an oil spill were to affect Sentosa Island, our beaches and those assets located along the island's coastline would be directly impacted. Siloso headlands falls within close proximity of Siloso beach.                                                        |
| 27  | Changi Beach Park (includes seawalls, revetments and Sungei Changi)                          | Park and Recreation Areas                                                  | NParks, NEA, BCA, PUB | 2                             | 1                  | 1                      | 3                   | 7              | Closest beach park to Pulau Ubin (Chek Jawa). Public recreational beach will have to be closed and cleaned up. High tide can bring oil inland up Sungei Changi.                                                                                                    |
| 28  | Pulau Ubin (Nordin Beach, Maman Jelutong Beach, including main jetty, revetment and seawall) | Park and Recreation Areas                                                  | NEA, BCA              | 2                             | 1                  | 1                      | 3                   | 7              | Public recreational beach will have to be closed and cleaned up                                                                                                                                                                                                    |
| 29  | Pasir Ris Park (include Sungei Api Api and breakwaters)                                      | Park and Recreation Areas                                                  | NParks, NEA, BCA      | 2                             | 1                  | 1                      | 3                   | 7              | Public recreational beach will have to be closed and cleaned up                                                                                                                                                                                                    |
| 30  | Punggol Beach (include seawall)                                                              | Park and Recreation Areas                                                  | NEA, BCA              | 2                             | 1                  | 1                      | 3                   | 7              | Public recreational beach will have to be closed and cleaned up                                                                                                                                                                                                    |
| 31  | Sembawang Park (include jetty and revetment)                                                 | Park and Recreation Areas                                                  | NParks, NEA, BCA      | 2                             | 1                  | 1                      | 3                   | 7              | Public recreational beach will have to be closed and cleaned up                                                                                                                                                                                                    |
| 32  | West Coast Park (include revetment, look-out deck and sluice gate)                           | Park and Recreation Areas                                                  | NParks, NEA, BCA      | 2                             | 1                  | 1                      | 3                   | 7              | Public recreational beach will have to be closed and cleaned up                                                                                                                                                                                                    |
| 33  | Woodlands Waterfront (include jetty and seawall)                                             | Park and Recreation Areas                                                  | NParks, BCA           | 2                             | 1                  | 1                      | 3                   | 7              | Built by URA but managed by NParks. Long jetty that runs parallel to the coastline. Popular for recreational fishing                                                                                                                                               |
| 34  | Labrador Nature Reserve (with Berlayar Creek, including jetty and revetment)                 | Park and Recreation Areas; Sensitive Marine and Coastal Biodiversity Areas | NParks, BCA, PUB      | 1                             | 1                  | 2                      | 3                   | 7              | One of 2 coastal nature reserves. Only reserve on mainland Singapore with a rocky shore habitat. Berlayar Creek has a substantial linear patch of mangroves that is the only one on southern mainland Singapore. High tide can bring oil inland up Berlayar Creek. |
| 35  | Sungei Buloh Wetland Reserve                                                                 | Sensitive Marine and Coastal Biodiversity Areas                            | NParks                | 1                             | 1                  | 2                      | 3                   | 7              | One of 2 coastal nature reserves.                                                                                                                                                                                                                                  |
| 36  | Chek Jawa (includes jetty)                                                                   | Sensitive Marine and Coastal Biodiversity Areas                            | NParks, BCA           | 1                             | 1                  | 2                      | 3                   | 7              | Rich coastal habitat on the eastern edge of Pulau Ubin                                                                                                                                                                                                             |
| 37  | Arthur Rd Outlet Drain                                                                       | Bays & Basins                                                              | PUB                   | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                                                                                                                                                                                    |

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|-----|-------------------------------------------------------|----------------------------------------------|----------------------|-------------------------------|--------------------|------------------------|---------------------|----------------|--------------------------------------------------------------------------------------------------------------|
|     | NAME                                                  | TYPE                                         |                      |                               |                    |                        |                     |                |                                                                                                              |
| 38  | Bayshore Park Outlet Drain                            | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              | High tide can bring oil inland to park & residential areas (Tanjong Katong, Telok Kurau, Siglap, Bedok area) |
| 39  | Bedok Canal                                           | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 40  | Joo Chiat Rd Outlet Drain                             | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 41  | Lucky Heights Outlet Drain                            | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 42  | Opera Estate Outlet Drain                             | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 43  | Parbury Avenue Outlet Drain                           | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 44  | Siglap Canal                                          | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 45  | Tanjong Katong Rd Outlet Drain                        | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              | High tide can bring oil inland to park & residential areas (Clementi, West Coast area)                       |
| 46  | Clementi West Coast Outlet Drain                      | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 47  | Heng Mui Keng Outlet Drain                            | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 48  | Science Park II Outlet Drain                          | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 49  | Sungei Pandan Kechil                                  | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 50  | West Coast Crescent Outlet Drain                      | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              | High tide can bring oil inland to park & residential areas (Pasir Ris, Tampines area)                        |
| 51  | PA Camp Outlet Drain                                  | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 52  | Sungei Api Api                                        | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 53  | Sungei Tampines                                       | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              | High tide can bring oil inland to residential areas (Sembawang, Yishun area)                                 |
| 54  | Sungei Sembawang                                      | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 55  | Sungei Khatib Bongsu                                  | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 56  | Sembawang Outlet Drain No. 2                          | Bays & Basins                                | PUB                  | 2                             | 1                  | 1                      | 3                   | 7              |                                                                                                              |
| 57  | Sentosa Cove                                          | Bays & Basins;<br>Marinas & Sea Sports Clubs | SDC                  | 2                             | 1                  | 1                      | 3                   | 7              | Oil will affect residential area (Sentosa Cove)<br><b>[Note: Plays host to major events on occasion]</b>     |
| 58  | Cruise Bay (HarbourFront terminal, residential areas) | Bays & Basins                                | MPA                  | 2                             | 1                  | 1                      | 3                   | 7              | Oil will affect residential area (Keppel Bay)                                                                |
| 59  | Brani Base                                            | Government Installations/<br>Project Areas   | PCG                  | 1                             | 1                  | 1                      | 1                   | 4              | Operation of water-cooled boats will be affected by oil-polluted waters.                                     |
| 60  | Gul Base                                              | Government Installations/<br>Project Areas   | PCG                  | 1                             | 1                  | 1                      | 1                   | 4              |                                                                                                              |